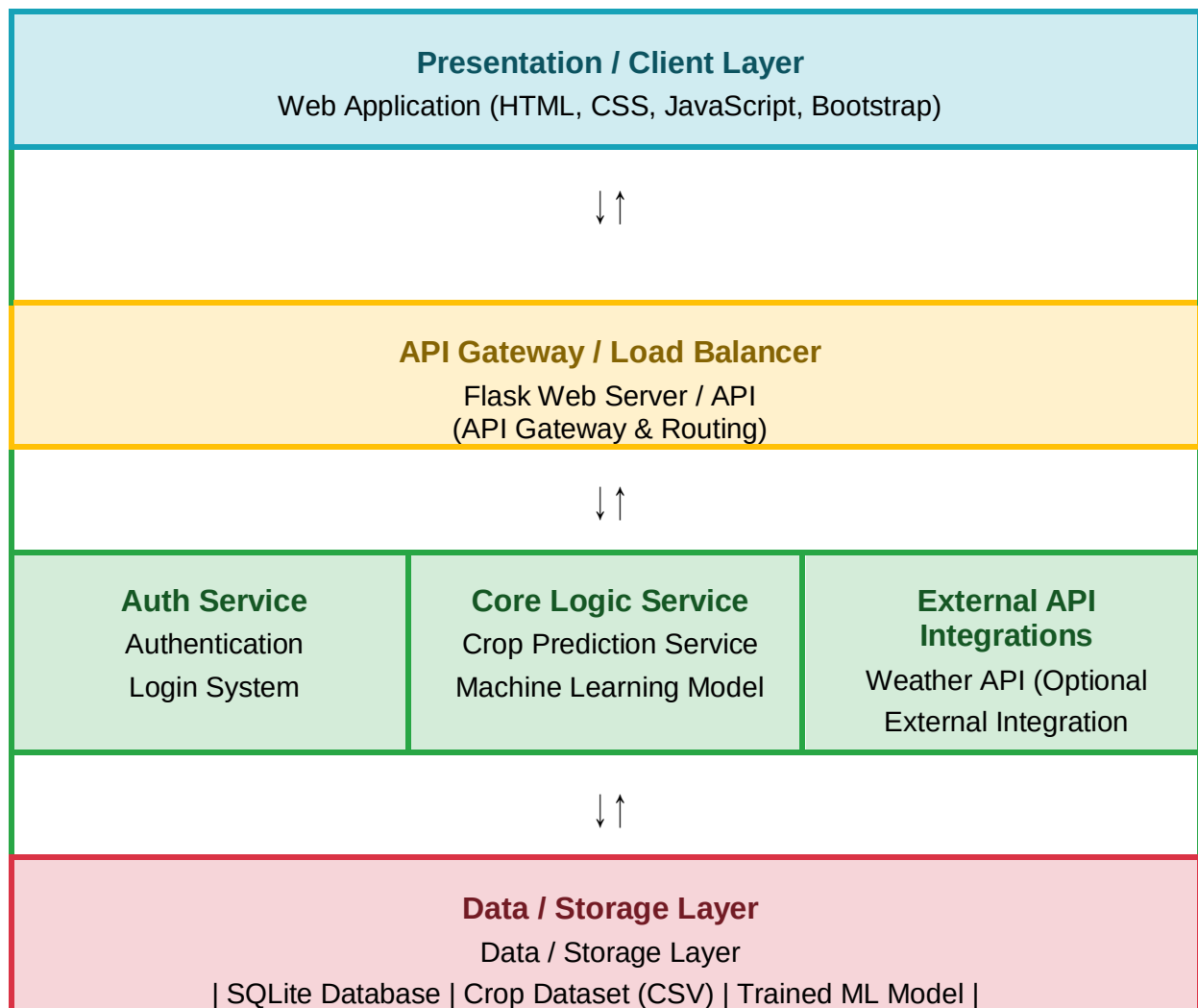


Solution Architecture

Date	5 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides the user interface where farmers register, log in, enter soil and weather details, and view crop prediction results.	HTML5, CSS3, JavaScript, Bootstrap
<i>API Gateway</i>	Receives user requests, routes them to the appropriate backend services, and returns prediction results to the frontend.	Flask
<i>Microservices</i>	Handles user authentication, processes input data, executes the Machine Learning model, and generates crop recommendations.	Python, Scikit-learn, Pandas, NumPy, Joblib
<i>Database</i>	Stores user information, login credentials, crop prediction history, and accesses the crop dataset and trained ML model.	SQLite, CSV Database]